

PROJECT CONTINUITY SUMMARY

1. Introduction

The Loan Prediction Analysis project was initiated in week 2 with the objective of understanding and preparing a loan application dataset for data-driven analysis and subsequent predictive modelling. Week 3 continued from the work completed during week 2 by extending the analysis into advanced exploratory data analysis, statistical analysis, feature engineering, feature evaluation and final dataset refinement. The purpose was to transform the prepared dataset into a reliable machine-learning ready dataset for predictive modelling in week 4.

2. Week 2 work completed

During week 2, the loan prediction dataset was initially explored and prepared for analysis. The dataset contained 614 observations and 13 variables, with Loan_Status representing the loan approval outcome.

The main week 2 activities included:

- Understanding the business problem and analytical objective,
- Inspecting the structure and variables of the dataset,
- Identifying missing values,
- Examining numerical and categorical variables,
- Performing initial exploratory data analysis,
- Cleaning the dataset,
- Preparing the data for machine-learning applications,
- Creating clean and machine-learning ready datasets.

Missing observations were identified in variables including Gender, Married, Dependents, Self_Employed, LoanAmount, Loan_Amount_Term and Credit_History.

3. Continuity into week 3

Week 3 did not repeat the complete week 2 cleaning and preprocessing process. Instead, the analysis progressed from the prepared loan-prediction workflow and focused on deeper investigation and refinement.

The week 3 analysis included:

- Advanced Exploratory Data Analysis (EDA),

- Numerical variable analysis,
- Categorical variable analysis,
- Correlation analysis,
- Multivariate analysis,
- Group comparisons,
- Statistical analysis,
- Feature engineering,
- Feature evaluation and selection,
- Dataset refinement,
- Categorical variable encoding,
- Feature scaling,
- Final modelling dataset preparation.

This progression allowed the project to move from basic data preparation towards evidence-based feature selection and machine-learning preparation.

4. Advanced exploratory data analysis

Advanced EDA was conducted to investigate patterns and relationships that were not fully explored during the initial analysis. Numerical variables were examined using distribution plots, histograms and boxplots. Categorical variables were examined using frequency distributions and grouped comparisons.

Multivariate analysis investigated relationships involving multiple variables, including:

- Applicant income, loan amount and loan status,
- Credit history, applicant income and loan status,
- Property area, loan amount and loan status.

Correlation analysis was also conducted to investigate relationships between numerical variables.

5. Statistical analysis

Statistical tests were conducted to determine whether observed relationships in the dataset were supported by statistical evidence. The analysis included correlation analysis, chi-square testing and the Mann-Whitney U test. Statistically significant associations were identified for variables including Married, Education and

Property_Area when compared with Loan_Status. CreditRiskCategory also demonstrated a strong statistical relationship with Loan_Status. However, this variable was subsequently removed because it was derived from Credit_History and therefore represented redundant information.

6. Feature engineering

Several meaningful features were developed during week 3 to provide additional information for analysis and modelling.

These included:

- TotalIncome
- IncomeBand
- DebtToIncomeRatio
- FamilySizeGroup
- CreditRiskCategory

The engineered features were evaluated based on their analytical usefulness and potential redundancy. CreditRiskCategory was ultimately excluded from the final modelling dataset because it duplicated information already represented by Credit_History.

7. Dataset Refinement

The week 3 findings were used to refine the dataset for machine-learning purposes. Loan_ID was excluded because it is an identifier and does not provide meaningful predictive information. CreditRiskCategory was also excluded because it was derived from Credit_History. The remaining variables were processed through categorical encoding and numerical feature scaling where applicable.

The final validation confirmed:

- Dataset shape: 614 rows × 23 columns,
- Predictor variables: 22,
- Target variable: 1,
- Total missing values: 0,
- Loan_Status missing values: 0.

The target variable contains:

- 422 approved applications,

- 192 rejected applications.

8. Outcome of week 3

The main outcome of week 3 is a refined, machine-learning ready dataset that incorporates the findings from advanced EDA, statistical analysis and feature evaluation. The final modelling dataset contains 23 variables, consisting of 22 predictor variables and the encoded Loan_Status target.

The refined dataset is saved as:

final_modelling_dataset.csv

9. Transition to week 4

The refined dataset will serve as the foundation for week 4 machine-learning activities.

The next stage will involve:

- Splitting the dataset into training and testing sets,
- Training classification models,
- Comparing model performance,
- Evaluating classification metrics,
- Investigating feature importance,
- Selecting an appropriate predictive model,
- Interpreting the model from a business perspective.

Therefore, week 3 provides the analytical and data preparation foundation required for predictive modelling in Week 4.